

Intramyocellular triglyceride content is a determinant of in vivo insulin resistance in humans: a ^1H - ^{13}C nuclear magnetic resonance spectroscopy as...

Insulin resistance is the best prediction factor for the clinical onset of type 2 diabetes. It was suggested that intramuscular triglyceride store may be a primary pathogenic factor for its development. To test this hypothesis, 14 young lean offspring of type 2 diabetic parents, a model of in vivo insulin resistance with increased risk to develop diabetes, and 14 healthy subjects matched for anthropomorphic parameters and life habits were studied with 1) euglycemic-hyperinsulinemic clamp to assess whole body insulin sensitivity, 2) localized ^1H nuclear magnetic resonance (NMR) spectroscopy of the soleus (higher content of fiber type I, insulin sensitive) and tibialis anterior (higher content of fiber type IIb, less insulin sensitive) muscles to assess intramyocellular triglyceride content, 3) ^{13}C NMR of the calf subcutaneous adipose tissue to assess composition in saturated/unsaturated carbons of triglyceride fatty acid chains, and 4) dual X-ray energy absorption to assess body composition. Offspring of diabetic parents, notwithstanding normal fat content and distribution, were characterized by insulin resistance and increased intramyocellular triglyceride content in the soleus ($P < 0.01$) but not in the tibialis anterior ($P = 0.19$), but showed a normal content of saturated/unsaturated carbons in the fatty acid chain of subcutaneous adipocytes. Stepwise regression analysis selected intramyocellular triglyceride soleus content and plasma free fatty acid levels as the main predictors of whole body insulin sensitivity. In conclusion, ^1H and ^{13}C NMR spectroscopy revealed intramyocellular abnormalities of lipid metabolism associated with whole body insulin resistance in subjects at high risk of developing diabetes, and might be useful tools for noninvasively monitoring these alterations in diabetes and prediabetic states.